LESSON 11

PATTERNS AND NUMBERS IN NATURE AND THE WORLD

Overview:

In this lesson, the students will be looking at different patterns and irregularities in the world and how Mathematics plays its role both in nature and human endeavor. They will see the appearance of these patterns in art, architecture, and nature.

Learning Objectives:

At the end of this lesson, the students can:

1. identify the next figure in a pattern.
2. recognize a pattern.
3. apply the concepts of pattern in real-life situations.
4. compute the n^{th} term of a Fibonacci sequence.
5. construct a Fibonacci spiral.

Materials Needed:

- Laptop/Cellphone
- Wifi/Data
- Video Lesson
- Module

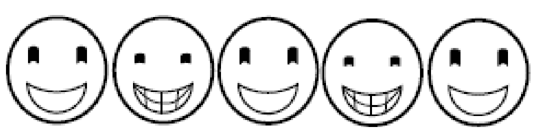
Durations: 4.5 hours

Learning Contents:

11.1 PATTERN

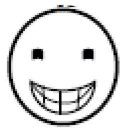
A **pattern** is an arrangement of things repeated in an orderly, recognizable fashion.

EXAMPLE: What is the next figure in the pattern below?



Solution.

It can be observed that the pattern above is made up of two smiling faces – one without teeth and one with teeth. The pattern begins with a toothless face, the two faces alternate. Logically, the face that should follow is



Numbers in a problem that are not given can be found by using established pattern.

EXAMPLE What number comes next in $1,3,5,7,9,$ ____?

Solution.

Looking at the given numbers, the sequence is increasing, with each term being two more than the previous term:

$$3 = 1 + 2,$$
$$5 = 3 + 2,$$
$$7 = 5 + 2,$$

and

$$9 = 7 + 2.$$

Therefore, the next term should be 11 since $11 = 9 + 2$.

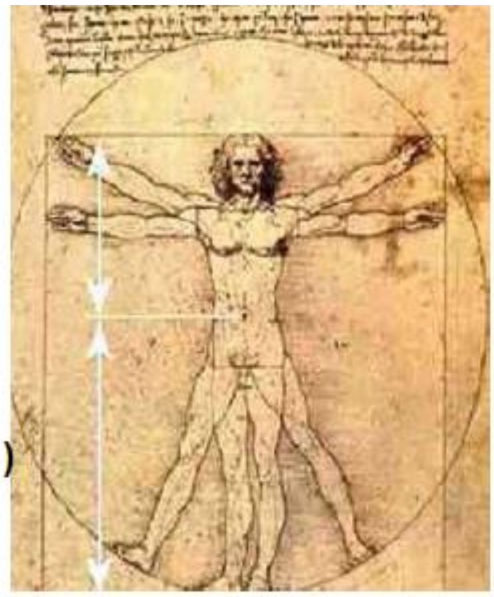
The appearance of patterns can also be seen in art, architecture, and nature. These patterns indicate a sense of structure and organization.

We are familiar with the sunflower. This flower demonstrates how nature works to optimize the available space. This arrangement allows the sunflower seeds to occupy the flower head in a way that maximizes their access to light and necessary nutrients.

Sunflower

Recall that **symmetry** indicates that you can draw an imaginary line (line of symmetry) across an object and the resulting parts are mirror images of each other. It is just like looking at a mirror and seeing how the left and the right sides of your face closely match. This is an example of a **bilateral symmetry**.

There are different types of symmetry depending on the number of sides or faces that are symmetrical. Some of these can be seen at the following images.



Leonardo da Vinci's Vitruvian Man showing the proportions and symmetry of the human body

Spiderwort with three-fold symmetry

This starfish has a five-fold symmetry

Notice also that different flowers vary with number of petals. Some flowers have three petals. However, flowers with five petals are said to be the most common.

Animals' external appearances also exhibited patterns. These patterns are believed to be governed by mathematical equations.Another wonderful nature's

Tiger

Hyena

designs are the structure and shape of a honeycomb and spiral patterns in whirlpool, shells of snails and other similar mollusks, and even in plants.

11.2 SEQUENCE

An ordered list of numbers such as $2,5,8,11,14,\cdots$ is called a **sequence**. The numbers in a sequence, that are separated by commas, are the **terms** of the sequence. In the above example, 2 is the first term, 5 is the second term, 8 is the third term, 11 is the fourth term, and 14 is the fifth term. The three dots " $\cdots$ " indicate that the sequence continues beyond 14, which is the last written.

It is customary to use the notation a_n to designate the $\boldsymbol{n^{th}}$ **term of a sequence**.

$$S_n = a_1, a_2, a_3, \cdots, a_n$$

where,

a_1 represents the first term of a sequence.
a_2 represents the second term of a sequence.
a_3 represents the third term of a sequence.
$\vdots$
a_n represents the n^{th} term of a sequence.

EXAMPLE What is the 10^{th} term in the sequence $2,6,12,20,30,\cdots,n^2+n,\cdots$?

Solution.

From the sequence,

$$a_1 = 2, a_2 = 6, a_3 = 12, a_4 = 20, a_s = 30, \text{ and } a_n = n^2 + n.$$

Hence, the 10^{th} term is $a_{10} = 10^2 + 10 = 100 + 10 = 110$.

Another way of examining the next term of a sequence and determining the n^{th} term formula is by constructing a difference table. A **difference table** shows the differences between successive terms of the sequence.

EXAMPLE Use a difference table to determine the next term of the previous sequence $2,5,8,11,14,$____.

Solution.

sequence:	2		5		8		11		14	⋯	
first difference		3		3		3		3			(1)

Thus, if we use the above difference table, we can predict the next term in the sequence is $14 + 3 = 7$.

EXAMPLE 12.5 Use a difference table to determine the next term of the sequence $5,14,27,44,65,\_\_\_$.

Solution.

sequence:	5		14		27		44		65	⋯	
first difference		9		13		17		21		⋯	(1)
second difference			4		4		4			⋯	(2)

In this difference table, the first differences (row 1) of the sequence are not all the same. In such situation, it is helpful to compute the successive differences of the previous differences until having all differences of the same constant. In the above example, the second differences are all 4. If the pattern continues, then a 4

sequence:	5		14		27		44		65		90 ⋯		
first difference		9		13		17		21		25		⋯	(1)
second difference			4		4		4		4			⋯	(2)

would also be added to the first difference, 21 to produce the second difference, 25. That is, $21 + 4 = 25$. We then add this difference to the fifth term of the sequence, 65, to predict that 90 is the next term of the sequence as shown below.

EXAMPLE Analyze the given sequence for its rule and identify the next three terms.

1. $1,10,100,1000$
2. $2,5,9,14,20$
3. $1,1,2,3,5,8,13$

Solution.

1. Looking at the set of numbers, each term is a power of 10: $10^0 = 1$, $10^1 = 10$, $10^2 = 100$, and $10^3 = 1000$. Following this rule, the next three terms are $10^4 = 10{,}000$, $10^5 = 100{,}000$, and $10^6 = 1{,}000{,}000$.

2. The difference between the first and second terms (2 and 5) is 3. The difference between the first and second terms (5 and 9) is 4. The difference between the first and second terms (9 and 14) is 5. The difference between the first and second terms (14 and 20) is 6. Continuing this pattern, it can be deduced that to obtain the next three terms we should add 7, 8, and 9, respectively, to the current term. Hence the next three terms are $20 + 7 = 27, 27 + 8 = 35$, and $35 + 9 = 44$.

3. Note that starting with 0 and 1, the succeeding terms in the sequence can be generated by adding the two previous terms.

$0 + 1$	$=$	1	0, 1, 1
$1 + 1$	$=$	2	0, 1, 1, 2
$1 + 2$	$=$	3	0, 1, 1, 2, 3
$2 + 3$	$=$	5	0, 1, 1, 2, 3, 5
$3 + 5$	$=$	8	0, 1, 1, 2, 3, 5, 8
$5 + 8$	$=$	13	0, 1, 1, 2, 3, 5, 8, 13
$8 + 13$	$=$	21	0, 1, 1, 2, 3, 5, 8, 13, 21
$13 + 21$	$=$	34	0, 1, 1, 2, 3, 5, 8, 13, 21, 34
$21 + 34$	$=$	55	0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55

Hence, the next three terms are 21, 34, and 55.

The previous example is of great historical interest - the **Fibonacci sequence**

$$1, 1, 2, 3, 5, 8, 13, 21, 34, 55, \ldots,$$

in which each term is the sum of the two preceding terms; for example, $55 = 21 + 34$. It is quite difficult for one to predict the 15^{th} term without listing all the previous ones. When we represent a sequence by describing the relationship between its successive terms, we are representing the sequence **recursively.**

11.3 THE FIBONACCI NUMBERS

Let F_n be the n^{th} number in a Fibonacci sequence, then the numbers in the Fibonaccai sequence are given by

$F_1 = 1$, $F_2 = 1$ and $F_n = F_{n-1} + F_{n-2}$ for $n \geq 3$.

EXAMPLE 12.7 Find the 10^{th} Fibonacci number sing the previous definition.

Solution. The 9 Fibonacci number are $1,1,2,3,5,8,13,21,34$, and 55. Hence,

$$F_{10} = F_9 + F_8 = 55 + 34 = 89.$$

GOLDEN RATIO 12.4

The ratios of the successive Fibonacci numbers approach the number (Phi), also known as the Golden Ratio. This is approximately equal to 1.618. By

$\frac{1}{1} = 1.0000$ $\quad \frac{5}{3} = 1.6667$ $\quad \frac{21}{13} = 1.6154$ $\quad \frac{89}{55} = 1.6182$

$\frac{2}{1} = 2.0000$ $\quad \frac{8}{5} = 1.6000$ $\quad \frac{34}{21} = 1.6190$

$\frac{3}{2} = 1.5000$ $\quad \frac{13}{8} = 1.6250$ $\quad \frac{55}{34} = 1.6177$

considering the Fibonacci sequence $1,1,2,3,5,8,13$

THE n^{th} –TERM FORMULA

The n^{th} – term formula is another way to describe a sequence by giving a formula for the n^{th} – term a_n.

EXAMPLE 12.8 What is the 6^{th} term in a sequence define by $a_n = 3n^2 + n$?

Solution.

Since $n = 6$, then

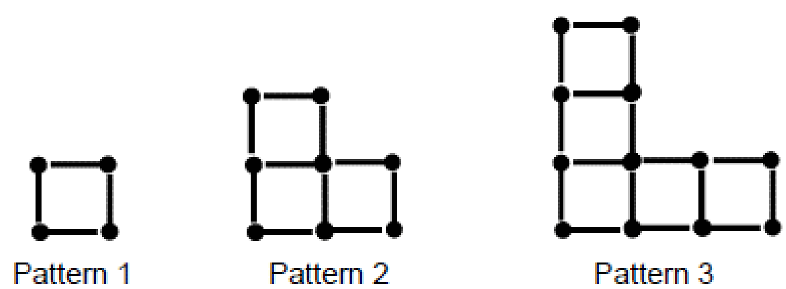
$a_6 = 3(6)^2 + 6$

$= 3(36) + 6$

$= 108 + 6$

$a_6 = 114$

EXAMPLE 12.9 How many match sticks will be needed for pattern 7?



Solution.

Patterns 1,2, and 3 have 4,10, and 16 match sticks respectively. The sequence 4, 10, 16 has a common difference of 6 and on the figure, as we move to the next pattern, we only add 3 match sticks on both upper and right sides of the base square instead of 4 match sticks. That should be 8 match sticks in all. Thus, we less two match sticks on every succeeding pattern. Hence, the *n*th-term formula of the pattern is given by

$$a_n = 6n - 2$$

To find the 7^{th} term, we have

$$a_n = 6n - 2$$
$$a_7 = 6(7) - 2$$
$$= 42 - 2$$
$$= 40$$

Therefore, 40 match sticks are needed for pattern 7.

Learning Activity:

A. Pattern Block Activity

Students will use a pattern block and work in pairs to construct a pattern. They will extend a pattern and state a rule for the pattern. Seven (7) random pairs are asked to present their works.

B. Creating the Fibonacci Spiral

Materials

1. Sheet of quarter-inch grid paper
2. Ruler
3. Compass

Follow the directions and watch the spiral emerge. (Each successive square will have one edge with a length the sum of the two squares immediately preceding it.)

1. Draw a 1-inch square. Draw a second 1-inch square to its left, making sure the squares touch.

2. Draw a 2-inch square above the two 1-inch squares, touching the lower squares.
3. Draw a 3-inch square to the right of the three existing squares; its left side should touch the other squares.
4. Draw a 5-inch square below these squares.
5. Draw an 8-inch square to the left of the existing five squares.
6. Draw a 13-inch square above the six squares.
7. Use a compass to complete the drawing. Within each square of your drawing, draw an arc from one corner to the opposite corner. (Each arc will have a radius equal to the length of one side of its square.) Connect each arc to the next. To begin, place your pencil in the upper-right corner of the original 1-inch square and draw an arc toward the lower-left corner. In the second square, draw an arc from that point (the lower-right corner) to the upper-left corner of the second square. Continue drawing arcs in each square, starting each arc at the point where the last one ended.
8. You will create a logarithmic spiral. What forms in nature reflect this shape?

Learning Evaluation:

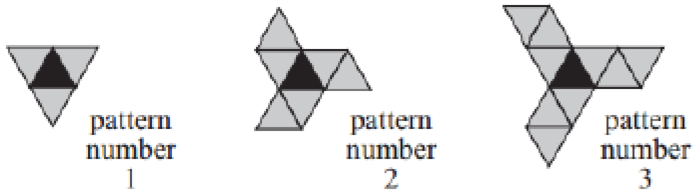
A. Determine what comes next in the given patterns.
   1. $A, C, E, G, I,$ _____
   2. $15,10,14,10,13,10,$ _____
   3. $3,6,12,24,48,96,$ _____
   4. $27,30,33,36,39,$ _____
   5. $41, 39, 37, 35, 35$_____

B. Let F_n be the n^{th} term of the Fibonacci sequence, with the $F_1 = 1$, $F_2 = 1$, $F_3 = 2$, and so on.

   1. Find F_8.
   2. Find F_{19}.
   3. If $F_{22} = 17{,}711$ and $F_{24} = 46{,}368$, what is F_{23}.
   4. Evaluate $F_1 + F_2 + F_3$.
   5. If you have a wooden board that is 0.75 meter wide, how long should you cut it such that the Golden ratio is observed? Use 1.618 as the value of the golden ratio.

C. Find the n^{th} –term formula of the given sequence and find the next term using the n^{th} - term formula you have.

1. $4, 7, 10\ , 13\ , 16, \cdots$
2. $5, 13, 21, 29, 37, \cdots$
3.



References:

Aufman, R. et. al. (2018). *Mathematics in the Modern World*. Rex Book Store Inc. Sampaloc, Manila, Philippines.

Johnstone, P. (1987). *Mathematics, Logic, Categories, and Sets*. Cambridge University Press.